Evaluate the following Vandermonde determinants:

$$\text{(a)} \begin{vmatrix} 1 & 2 & 2^2 \\ 1 & 3 & 3^2 \\ 1 & 6 & 6^2 \end{vmatrix}; \quad \text{(b)} \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}.$$

Vandermonde Determinant Formula

For distinct scalars x_1, x_2, x_3, the Vandermonde determinant

$$V = \begin{vmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{vmatrix} = \prod_{1 \le i < j \le 3} (x_j - x_i) = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2).$$

This follows from row reduction: subtract row 1 from rows 2,3; factor $(x_2 - x_1)$, $(x_3 - x_1)$ from those rows; the remaining 2×2 Vandermonde gives $(x_3 - x_2)$.

Solution for Part (a)

The matrix entries identify $x_1 = 1$, $x_2 = 2$, $x_3 = 6$ (row i: $[1, x_i, x_i^2]$ where $2^2 = 4$, $3^2 = 9$, $6^2 = 36$).

Apply formula:

$$\det = (2 - 1)(6 - 1)(6 - 2) = (1)(5)(4) = 20.$$

Verification by cofactor expansion along first row:

$$\det = 1 \begin{vmatrix} 3 & 9 \\ 6 & 36 \end{vmatrix} - 2 \begin{vmatrix} 1 & 9 \\ 1 & 36 \end{vmatrix} + 4 \begin{vmatrix} 1 & 3 \\ 1 & 6 \end{vmatrix}.$$

Compute minors:

$$\begin{vmatrix} 3 & 9 \\ 6 & 36 \end{vmatrix} = 3 \cdot 36 - 9 \cdot 6 = 108 - 54 = 54,$$

$$\begin{vmatrix} 1 & 9 \\ 1 & 36 \end{vmatrix} = 36 - 9 = 27,$$

$$\begin{vmatrix} 1 & 3 \\ 1 & 6 \end{vmatrix} = 6 - 3 = 3.$$

Thus: $1 \cdot 54 - 2 \cdot 27 + 4 \cdot 3 = 54 - 54 + 12 = 12$.

(Note: Direct computation gives 12; formula assumes standard row labeling but matrix uses $x_2 = 3$? Wait—row2 $[1, 3, 9]$ implies $x_2 = 3$, not 2 in col2 row1. Actually points are $x_1 = 1$ (implied), $x_2 = 3$, $x_3 = 6$? No, col2 gives the x_i: 2,3,6.)

Correct points: row1 $x = 2$, row2 $x = 3$, row3 $x = 6$. First column all 1's (x^0).

Formula: $(3 - 2)(6 - 2)(6 - 3) = (1)(4)(3) = 12$.

Solution for Part (b)

Points $x_1 = a$, $x_2 = b$, $x_3 = c$ (standard labeling).

$$\det = (b-a)(c-a)(c-b).$$

Verification by expansion:

$$\det = 1\begin{vmatrix} b & b^2 \\ c & c^2 \end{vmatrix} - a\begin{vmatrix} 1 & b^2 \\ 1 & c^2 \end{vmatrix} + a^2\begin{vmatrix} 1 & b \\ 1 & c \end{vmatrix}.$$

Minors:

$$b \cdot c^2 - b^2 \cdot c = bc(c-b),$$
$$c^2 - b^2 = (c-b)(c+b),$$
$$c - b = c - b.$$

Thus:

$$bc(c-b) - a(c-b)(c+b) + a^2(c-b) = (c-b)[bc - a(c+b) + a^2] = (c-b)(a^2 - ab - ac + bc).$$

Factor further: $a^2 - ab - ac + bc = a(a-b-c) + bc$, or recognize $(b-a)(c-a)$.
Full symmetric expansion confirms $(b-a)(c-a)(c-b)$.
Final answers: (a) 12*; (b)* $(b-a)(c-a)(c-b)$.